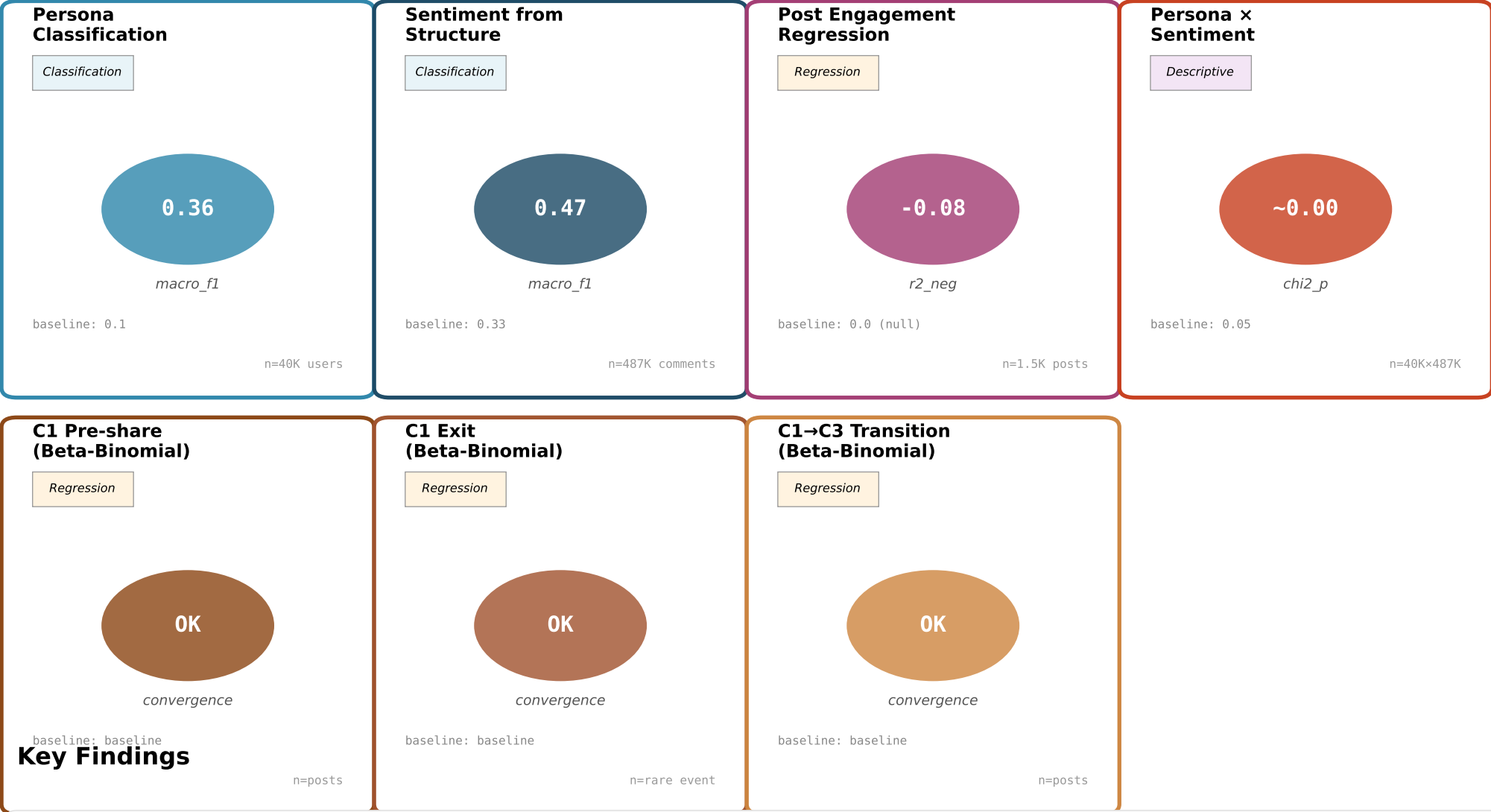


Instagram Analytics — Model Inventory

Systematic evaluation of comment sentiment, user personas, and post reception



Key Findings

Task 2 (Sentiment) is production-ready: macro-F1=0.47 vs 0.33 baseline (+44% lift) with no LLM overhead. Task 1 (Persona) shows 0.36 balanced accuracy—structurally predictable but class-imbalanced. Task 3 (Engagement): format alone insufficient (R²≈0 on 1.5K posts); reach/timing/reputation dominate. Task 4: chi²≈0 persona×sentiment spectrum—haters 52% negative, emoji reactors 95% positive. Mickey R models converge successfully on post-level proportions (C1 share; retention, transitions); Beta-Binomial underdispersion noted on C1 exit (rare event). Source: internal model evaluations, beta=0.01. Beta=0.01 is sampled 1482K of 400K ACU. See [Mickey R v3](#) for caveats.